Steps: 1.- Derivo el polinomio buscado

2.- Se integra indefinidamente y aparece la cte: c.

3.- Vuelvo a integrar entre 0 y 1 para obtener constante

Propiedades a utilizar:

1) $B_0^{\wedge}(x) = 1$

2) $D B_\nu^{\wedge}(x) = \nu B_{\nu-1}^{\wedge}(x)$

3) $\int_0^1 B_\nu(x)\, dx = 0$; si $\nu \neq 0$

Seed: $B_0(x) = 1$

$$D B_1(x) = B_0(x) = 1$$

$$\int D B_1(x)\, dx = \int 1\, dx$$

$$B_1(x) = x + c$$

$$\int_0^1 B_1(x)\, dx = \left. \frac{x^2}{2} + cx \right|_0^1$$

$$0 = \frac{1}{2} + c$$

$$-\frac{1}{2} = c$$

Luego $\boxed{B_1(x) = x - \frac{1}{2}}$

Busco $B_2(x)$

Teniendo en cuenta que $B_1(x) = x - \frac{1}{2}$

$$D B_2(x) = 2 B_1(x)$$

$$\int D B_2(x)\, dx = 2 \int \left(x - \frac{1}{2}\right) dx$$

$$B_2(x) = 2\left(\frac{x^2}{2} - \frac{x}{2}\right) + c$$

$$\int_0^1 B_2(x)\, dx = \int_0^1 (x^2 - x + c)\, dx$$

$$0 = \frac{x^3}{3} - \frac{x^2}{2} + c \Big|_0^1$$

$$c = -\frac{1}{3} + \frac{1}{2} = \frac{1}{6}$$

Luego

$$\boxed{B_2(x) = x^2 - x + \frac{1}{6}}$$

Busco $B_3(x)$

$$D B_3(x) = 3 B_2(x)$$

$$\int D B_3(x)\, dx = \int \left(3x^2 - 3x + \frac{1}{2}\right) dx$$

$$B_3(x) = x^3 - \frac{3}{2}x^2 + \frac{x}{2} + c$$

$$\int_0^1 B_3(x)\,dx = \frac{x^4}{4} - \frac{x^3}{2} + \frac{x^2}{4} + cx \Big|_0^1$$

$$0 = \frac{1}{4} - \frac{1}{2} + \frac{1}{4} + c$$

$$c = 0$$

Luego $\boxed{B_3(x) = x^3 - \frac{3}{2}x^2 + \frac{x}{2}}$

Busco $B_4(x)$

$$D B_4(x) = 4 B_3(x)$$

$$\int D(B_4(x))\,dx = \int \left(4x^3 - 6x^2 + 2x\right) dx$$

$$B_4(x) = x^4 - 2x^3 + x^2 + c$$

$$\int_0^1 B_4(x)\,dx = \frac{x^5}{5} - \frac{x^4}{2} + \frac{x^3}{3} + cx \Big|_0^1$$

$$0 = \frac{1}{5} - \frac{1}{2} + \frac{1}{3} + c$$

$$c = \frac{-6+15-10}{30} = -\frac{1}{30}$$

$$\boxed{B_4(x) = x^4 - 2x^3 + x^2 - \frac{1}{30}}$$